WIKIPEDIA* 25 years of the free encyclopedia Machine learning Machine learning (ML) is a field of study in artificial intelligence concerned with the development and study of statistical orithms that can learn from data and generalize to unseen data, and thus perform tasks without being explicitly 1 Statistics and mathematical optimisation methods compose the foundations of machine learning. Data mining is a related field of study, focusing on exploratory data analysis (EDA) through unsupervised learning.(3JL4.J From a theoretical viewpoint, probably approximately correct learning provides a mathematical and statistical framework for describing machine learning. Most traditional machine learning and deep learning algorithms can be described as empirical risk minimisation under this framework. Advances in the field of deep learning have allowed neural networks a class of statistical algorithms, to surpass many previous machine learning approaches in performance. The resulting dominance of machine learning within the field of AI from the late 2000s and early 2010s onwards was such that the terms "machine learning" and "artificial intelligence" are now often used interchangeably, even though they are not Synonymous History The term machine learning was coined in 1959 by Arthur Samuel, an IBM employee and pioneer in the field of computer gaming and artificial intelligence The synonym self-teaching computers was also used during this time period. The earliest machine learning program was introduced in the 1950s, when Samuel invented a computer program that calculated the chance of winning in checkers for each side, but the history of machine learning is rooted in decades of efforts to study human cognitive processes-O In 1949, Canadian psychologist Donald Hebb published the book Ihe anization Behavior, in which he introduced a theoretical neural structure formed by certain interactions among nerve cells. 11 Hebbian theow of neuron interaction set the groundwork for how many machine learning algorithms work, with connected artificial neurons changing the strength of their connections based on data.@PJ Other researchers who have studied human cognitive Systems contributed to the modern machine learning technologies as well, including Walter Pitts and Warren McCulloch, who proposed the first mathematical model of neural networks including algorithms that mirror human thought processes. 10 By the early 1960s, an experimental "learning machine" with punched tape memon', called Cybertron, had been developed by Company to analyse sonar signals, electrocardiograms, and speech patterns using rudimentary It was repetitively "trained" by a human operator/teacher to recognise patterns and equipped with a "goof' button to cause it to reevaluate incorrect decisions 12 A representative book on research into machine learning during the 1960s was Nils Nilsson's book "Learning Machines", dealing mostly with machine learning for pattern classification.L.!3J Interest related to pattern recognition continued into the 1970s, as described by Duda and Hart in 1973.04) In 1981, a report was given on using teaching strategies so that an artificial neural network learns to recognise 40 characters (26 letters, 10 digits, and 4 special symbols) from a computer terminal.05J Tom M. Mitchell provided a widely quoted, more formal definition of the algorithms studied in the

machine learning field: "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T, as measured by P, improves with experience This definition of the tasks in which machine learning is concerned is fundamentally operational rather than defining the field in cognitive terms. This follows Alan Turing's proposal in his paper "Computing Machinery and Intelligence", in which the question, "Can machines think?", is replaced by asking whether machines can convincingly imitate a human in its responses to human-posed questions. 1 18 In 2012, AlexNet, developed by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, achieved substantially improved results in the ImageNet image recognition competition, contributing to the wider adoption of deep neural networks.L!9J In 2013, Tom" Mikolov and colleagues introduced word2vec, techniques for efficiently learning distributed vector representations of words from large text corpora. [2011211 In 2014, Ian Goodfellow and colleagues introduced generative adversarial networks (GANs), a framework for training generative models through an adversarial process.[ggl In 2016, AlphaGo became the first computer program to defeat a professional human Go player without handicaps on a full-sized board, using deep neural networks and reinforcement learning In 2017, Ashish Vaswani and colleagues introduced the Transformer a neural network architecture based primarily on attention rather than recurrence or Relationships to other fields Artificial intelligence As a scientific endeavour, machine learning grew out of the quest for artificial intelligence (AI). In the early days of AI as an academic discipline, some researchers were interested in having machines learn from data. They attempted to approach the problem with various symbolic methods, as well as what were then termed "neural networks"; these were mostly and Other models that were later found to be reinventions of the generalised linear models of statistics.L?éJ Probabilistic reasoning was also employed, especially in automated medical diagnosis However, an increasing emphasis on the logical, knowledge-based approach caused a rift between AI and machine learning. Probabilistic systems were plagued by theoretical and practical problems of data acquisition and By 1980, expert systems had come to dominate AI, and statistics was out of Work on symbolic/knowledge-based learning continued within AI, leading to inductive logic programming (ILP), but the more statistical line of research was now outside the field of AI proper, in pattern recognition and information Neural network research was abandoned by Al and computer science around the same time. This subfield, termed "connectionism", was continued by researchers from other disciplines, including John Hopfield, David Rumelhart, and Geoffrey Hinton. Their main success came in the mid-1980s with the reinvention of backpropagation. : 25 Machine learning (ML), reorganised and recognised as its own field, started to flourish in the 1990s. The field changed its goal from achieving artificial intelligence to tackling solvable problems of a practical nature. It shifted focus away from the symbolic approaches it had inherited from AI, and toward methods and models borrowed from statistics, fuzzy logic, and Data compression Artificial Intelligence: Deep learning is a subset or machine

learning. which is itself a subset of artificial intelligenceJ?9.l There is a close connection between machine learning and compression. A system that predicts the posterior probabilities of a sequence given its entire history can be used for optimal data compression (by using arithmetic coding on the output distribution). Conversely, an optimal compressor can be used for prediction (by finding the symbol that compresses best, given the previous history). This equivalence has been used as a justification for using data compression as a benchmark for "general intelligence . An alternative view can show compression algorithms implicitly map strings into implicit feature space vectors, and compression-based similarity measures compute similarity within these feature spaces. For each compressor C(.) we define an associated vector space N, such that C(.) maps an input string x, corresponding to the vector norm I l. An exhaustive examination of the feature spaces underlying all compression algorithms is precluded by space; instead, feature vectors chooses to examine three representative lossless compression methods, LZW, 1177, and PPM.L3?J According to AIXI theory, a connection more directly explained in Hutter Prize the best possible compression of x is the smallest possible software that generates x. For example, in that model, a zip file's compressed size includes both the zip file and the unzipping software, since you can not unzip it without both, but there may be an even smaller combined form. Examples of Al-powered audio/video compression software include NVIDIA Maxine, AIVC.L33J Examples of software that can perform Al-powered image compression include Qpgggy, TensorFlow, MATLAB's Image Processing Toolbox (IPT) and High-Fidelity Generative Image Compression. In unsupervised machine learning, k-means clustering can be utilized to compress data by grouping similar data points into clusters. technique simplifies handling extensive datasets that lack predefined labels and finds widespread use in fields such as com ssion Data compression aims to reduce the size of data files, enhancing storage efficiency and speeding up data transmission. K-means clustering, an unsupervised machine learning algorithm, is employed to partition a dataset into a specified number of clusters, k, each represented by the centroid of its points. This process condenses extensive datasets into a more compact set of representative points. Particularly beneficial in and signal processing, k-means clustering aids in data reduction by replacing groups of data points with their centroids, thereby preserving the core information of the original data while significantly decreasing the required storage space. Large language models (LLMs) are also efficient lossless data compressors on some data sets, as demonstrated by DeepMind's research with the Chinchilla 70B model. Developed by DeepMind, Chinchilla 70B effectively compressed data, outperforming conventional methods such as Portable Network Graphics (PNG) for images and Free Lossless Audio Codec (FLAC) for audio. It achieved compression of image and audio data to 43-4% and 16.4% of their original sizes, respectively. There is, however, some reason to be concerned that the data set used for testing overlaps the LLM training data set, making it possible that the Chinchilla 70B model is only an effcient compression tool on data it has already been trained on.[37JL3Y) Data mining

Machine learning and data mining often employ the same methods and overlap significantly, but while machine learning focuses on prediction based on known properties learned from the training data, data mining focuses on the of previously unknown properties in the data (this is the analysis step of knowledge discovery in databases). Data mining uses many machine learning methods, but with different goals; on the other hand, machine learning also employs data mining methods as "unsupervised learning" or as a preprocessing step to improve learner accuracy. Much of the confusion between these two research communities comes from the basic assumptions they work with' in machine learning, performance is usually evaluated with respect to the ability to reproduce known knowledge, while in knowledge discovery and data mining (KDD) the key task is the discovery of previously unknown knowledge. Evaluated with respect to known knowledge, an uninformed (unsupervised) method will easily be outperformed by other supervised methods, while in a typical KDD task, supervised methods cannot be used due to the unavailability of training data.